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TOS²CA: The Thematic Observation Search, Segmentation, Collation and Analysis System

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Software

- Review
- Repository
- Archive

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Summary

Thematic Observation Search, Segmentation, Collation and Analysis (TOS²CA) is a user-driven, data-centric system that identifies, collates, statistically characterizes, and serves Earth science data relevant to a given phenomenon. Developed as a modular, Python-based, cloud-native software library, TOS²CA facilitates the collation and analysis of data from disparate sources and makes it possible for scientists to establish science traceability requirements, quantify detection thresholds, define uncertainty requirements, and establish data sufficiency to formulate truly innovative missions and research investigations.

Statement of need

TOS²CA arose from the research need to identify (detect and track) occurrences of physical phenomena based on a user's definition within given spatial and temporal bounds (see Figure 1). Once identified, the phenomena information can be used to curate a multi-variable data set for different analyses.

An example of a real-world research question that TOS²CA could address would be "What atmospheric conditions produce deep convective storms, somewhat intense convection, shallow convection, and no storm?". TOS²CA can answer this by delineating storms from the program of record (e.g., geostationary IR), curating condition variables (e.g., temperature, moisture, shear), and then finding the probability distribution function of the condition variables when any given storm is different from the probability distribution function when there is no storm.

State of the field

There are many tools that exist to subset, collect, harmonize, interpolate and plot Earth science data sets. For example, NASA has developed `Harmony` (National Aeronautics and Space Administration, n.d.-c) that can access and subset NASA Earth science data, `Earthdata Search` (National Aeronautics and Space Administration, n.d.-a) that can search and subset data, and `Giovanni` (National Aeronautics and Space Administration, n.d.-b) that can create visualizations. However, there is no known tool or software package available that (1) allows users to define their own phenomena in a more meaningful way than setting temporal and spatial bounds, (2) gathers and interpolates data to the bounds of the mask, and (3) performs the definition, curation, and analysis work in a systematic end-to-end fashion.

Software design

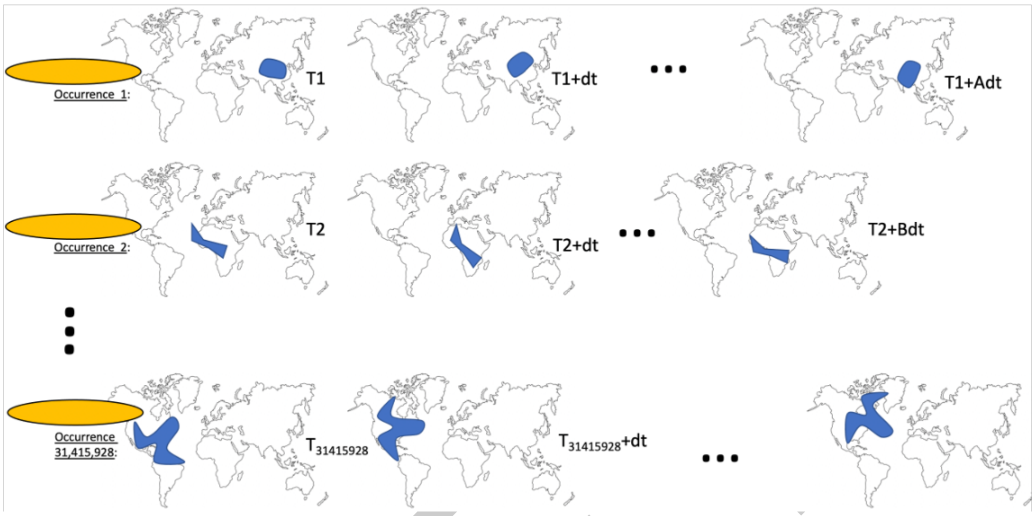

Figure 1: Illustration of a TOS^{2}CA a chronology of occurrences. TOS^{2}CA accepts a user specified defining condition and finds all phenomena occurrences in a selected data set that meet the condition. A netCDF-4 mask file is then generated with all the locations of each occurrence at each timestep. This allows users to track each phenomena spatially and temporally.

TOS^{2}CA was designed to operate in three stages:

1. **Phenomenon Definition (PhDef)**: The user supplies information on the data set and variable they want to use to search for phenomena, along with spatial bounds, temporal ranges, and a conditional. The data set will be evaluated by a user's chosen algorithm which will – on a pixel-by-pixel basis – identify and track phenomena to generate masks (see illustration in Figure 1).
2. **Data Curation**: With the masks generated in PhDef, the user selects which data sets and variables they want to collect for analysis. That data is then fit to the bounds of the phenomenon masks and interpolated to the temporal and spatial resolution of the data set that was used to generate the masks (see Figure 2).
3. **Visualization**: Interactive analysis tools use the data generated in Data Curation to compare multiple variables to each other (see Figure 3) in customizable plots and charts.

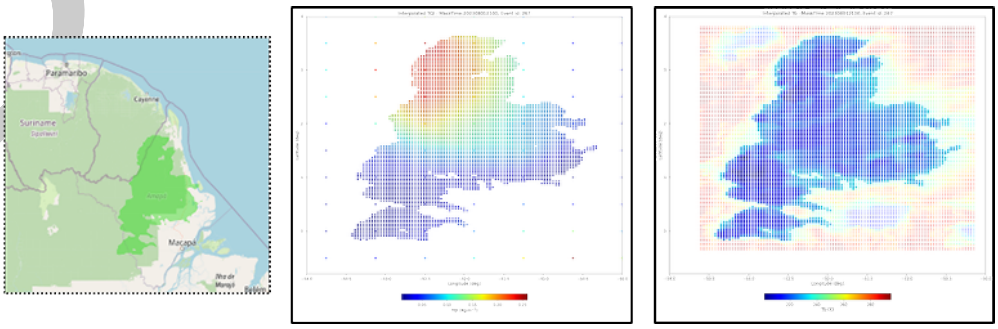

Figure 2: A mask of a phenomena at a specific timestep (left, in dark green). Data from *n* additional data sets can be curated and fit to the bounds of the mask. Here, data from NASA's Modern-Era Retrospective analysis for Research and Applications, Version 2 (MERRA-2) total precipitable ice water (middle) and GPM brightness temperature (right) have been fit to the bounds of this mask. The data are also interpolated to the resolution of the data set used to create the mask so that everything can be easily compared.

The TOS^{2}CA software library (Haddad et al., 2025) (Knosp et al., 2024) is modular and takes advantage of data and computing resources in the cloud (specifically Amazon Web Services or "AWS"). The software is hosted in the nasa-jpl GitHub organization as a collection of repositories:

- **tos2ca-anomaly-detection**: The main TOS^{2}CA repository that includes data readers and curators, algorithm connectors, utilities (e.g., interpolators, plotters), and database connectors. New data readers and curators can be easily dropped in to the `iolib` folder for use in the system (after updating the system's data dictionaries). Users can also drop in connectors for additional algorithms in the `utils` folder. This repository has branches with Jupyter Notebooks showing examples of how to use curated data.
- **tos2ca-fortracc-module**: A repository containing the ForTraCC Module (Vila et al., 2008) that tracks spatially and temporally connected clusters from a user definition in the PhDef stage.
- **tos2ca-aux-geoir**: A repository containing the AUX-GEOIR algorithm that spatially and temporally tracks convective storms in Global Precipitation Measurement Mission's Integrated Multi-satellitE Retrievals for GPM (IMERG) data in the PhDef stage.
- **tos2ca-data-dictionaries**: This repository contains JSON files with information about data that TOS^{2}CA accesses in the PhDef and Data Curation stages, such as variable names and data URLs/URIs.
- **tos2ca-user-interface**: This repository contains a web-based user interface and API architecture that allows users to submit and track jobs, and was desinged for a LAMP (Linux, Apache, MySQL, and PHP) configuration and with FastAPI, respectively.
- **tos2ca-data-access-server**: A repository containing web-based analysis tools that use the mask and curated data sets.
- **tos2ca-documentation**: This documentation repository contains a Docusaurus site that has information about the TOS^{2}CA project.
- **tos2ca-containerization**: A repository with examples of how to containerize the TOS^{2}CA code, running jobs in temporal chunks.

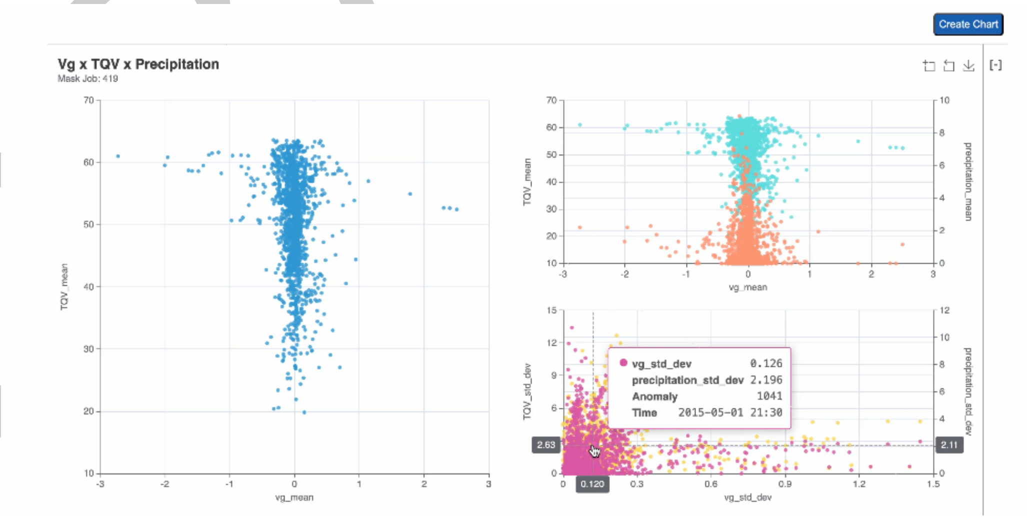

Figure 3: TOS^{2}CA analysis tools where users can jointly analyze multiple data variables curated from the system. Here we look see meridional geostrophic surface current (vg), total precipitable water vapor (TQV), and precipitation plotted together.

TOS^{2}CA was developed and tested in AWS. It can launch jobs serially on a standalone server (like an AWS Elastic Compute Cloud instance) or chunk jobs to be run in an embarrassingly parallel manner on an HPC (like AWS Fargate). We found that the majority of TOS2^{2}CA's performance is not limited by CPU speed, but rather memory, which is why the suggested way to run TOS^{2}CA is by chunking jobs and running the chunks in parallel. The team chose to deploy its AWS resources in the `us-west-2` AWS region, co-locating resources region-wise with NASA's Earthdata in the Cloud data sets to eliminate data egress fees. It was also developed for use with Level 3 (gridded) data sets, though the team has tested it with a Level 2 (swath) data set from the Advanced Scatterometer (ASCAT) missions. (An ASCAT reader and curator is included with the code but has not completed testing.)

Research impact statement

TOS^{2}CA was developed for NASA's INvestigation of Convective UpdraftS (INCUS) mission (Kim et al., 2023) (van den Heever et al., 2022). INCUS' objective is to answer, "Why do convective storms, heavy precipitation, and clouds occur exactly when and where they do?". INCUS will make observations of convective updrafts (the building blocks of convective storms) systematically globally over 2 years, then relate these data to the initial storm environment E, on one hand, and the storm outcomes (anvil cloud outcomes A and precipitation outcomes R) by producing the conditional distributions:

$p(\ CU \mid E\ )$, then $p(\ A \mid CU\ )$, $p(\ R \mid CU\ )$

The E, A, R observations will come from the program of record, and be used to:

- Delineate the storms from the program of record data for E, A, R
- Partition the E data according to environment type
- Describe each $p(\ CU \mid Ei\ )$

The conditional distributions

$p(\ CU \mid Ei\ )$, then $p(\ CU \mid Ai\ )$, $p(\ CU \mid Ri\ )$

can only be done after the INCUS observations are made (2026 – 2028). However, today we can ask – and need to answer – what it takes to describe:

$p(\ A \mid Ei\ )$ and $p(\ R \mid Ei\ )$ for $i = 1, \dots, N$

as that will determine how the partitioning should be done (and how big N should be). This requires delineating the storms and then curating the storm masks with E, A, R data to examine the correlations. TOS^{2}CA is designed to perform the majority of the work needed to provide this answer. Beyond INCUS, TOS^{2}CA can help answer similar questions for almost any type of physical phenomena that a user wants to define.

Additional details

More information on TOS^{2}CA can be found on its documentation website at: https://nasa-jpl.github.io/tos2ca-documentation/

AI usage disclosure

Generative AI tools were used for limited assistance during the development of this software, such as creating some minor functions and helping to resolve bugs. None were used in the writing and of this manuscript or the preparation of supporting materials.

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